

BCSE324L	FOUNDATIONS OF BLOCKCHAIN TECHNOLOGY			L	T	P	C
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Pre-requisite	NIL	Syllabus version					
		1.0					
Course Objectives							
1. To understand building blocks of Blockchain.							
2. To significance of Distributed Ledger Technology and Smart Contract.							
3. To exploit applications of Blockchain in real world scenarios and their impacts.							
Course Outcomes							
After completion of this course, the student shall be able to:							
1. Understand Blockchain ecosystem and its services in real world sceneries							
2. Apply and Analyze the requirement of Distributed Ledger Technology and Smart Contract							
3. Design and Demonstrate end-to-end decentralized applications							
4. Acquaint the protocol and assess their computational requirements							
Module:1	Foundations of Blockchain					7 hours	
Blockchain Architecture – Challenges – Applications – Blockchain Design Principles -The Blockchain Ecosystem - The consensus problem - Asynchronous Byzantine Agreement - AAP protocol and its analysis - peer-to-peer network – Abstract Models - GARAY model - RLA Model - Proof of Work (PoW) - Proof of Stake (PoS) based Chains - Hybrid models.							
Module:2	Distributed Ledger Technology					6 hours	
Origin of Ledgers – Types and Features of Distributed Ledger Technology (DLT) - Role of Consensus Mechanism - DLT Ecosystem - Distributed Ledger Implementations – Blockchain - Ethereum - Public and Private Ledgers – Registries – Ledgers - Practitioner Perspective: Keyless Technologies, Transparency as a Strategic Risk, Transparency as a Strategic Asset, Usage of Multiple IDs - Zero Knowledge Proofs - Implementation of Public and Private Blockchain							
Module:3	Smart Contracts					5 hours	
Anatomy of a Smart Contracts - Life Cycle - Usage Patterns - DLT-based smart contracts - Use Cases: Healthcare Industry and Property Transfer.							
Module:4	Decentralized Organization					5 hours	
Decentralization versus Distribution - Centralized-distributed (Ce-Di) organizations - Decentralized-distributed (De-Di) organizations - Decentralized Autonomous Organizations: Aragon, DAOstack, DAOhaus and Colony.							
Module:5	Types of Blockchain Ecosystem					7 hours	
One-Leader Ecosystem - Joint Venture or Consortia Ecosystems - Regulatory Blockchain Ecosystems - Components in Blockchain Ecosystem: Leaders, Core Group, Active Participants, Users, Third-Party Service Providers - Governance for Blockchain Ecosystems.							
Module:6	Blockchain Protocols					6 hours	
Ethereum tokens – Augur - Golem - Understanding Ethereum tokens - App Coins and Protocol Tokens - Blockchain Token Securities Law Framework - Token Economy - Token sale structure - Ethereum Subreddit.							
Module:7	High Performance Computing					7 hours	
Integrity of High Performance Systems - Data Provenance - Cluster Construction and Deployment - Mock Workload - Blockchain Software Evaluation - Blockchain storage of Integrity Data.							
Module:8	Contemporary Issues					2 hours	
	Total Lecture hours:					45 hours	
Text Book							
1.	Dhillon, V., Metcalf, D., and Hooper, M, Blockchain enabled applications, 2017, 1st						